

When Does Filtering Help You See a Break? Latent-State Diagnostics for Structural Change at Calibrated False-Alarm Rates

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Abstract

We ask when filtering a latent state helps detect structural change, under a protocol that makes the question answerable: every detector — state-aware or raw-data — is calibrated on matched null data-generating processes to the same false-alarm rate (5% per 500 observations), so detection rates and delays are directly comparable. The answer is a trichotomy — *no, yes, no* — across three break types. (i) Level shifts in a persistent state: *no* — a raw-data CUSUM dominates at every SNR, and we prove the innovation CUSUM is “fast or never,” a boundary condition of persistence ($\mu\infty$ sorts detection, Spearman 0.94). (ii) Observation-noise variance changes: *yes* — but the advantage is prewhitening, not the state estimate: the observable is exactly ARMA(1,1), so ARIMA-residual and Kalman-innovation whitening are provably the same filter. (iii) State-innovation (shock) variance changes — the Great-Moderation / crisis-volatility channel: *no*, a partial null. On the coarse $\times 3$ break, where all rungs detect above chance, a raw variance CUSUM matches or beats whitening at every SNR (0.72/0.96/0.96 vs ARIMA 0.26/0.79/1.00). The subtle $\times 1.5$ break, by contrast, leaves every rung near the false-alarm floor — whitening *fails to recover* it rather than raw detecting it. On real data (industrial production, GDP, Treasury yields, and unemployment) every alarm attributes to a second-moment feature, NBER association is significant (permutation $p = 0.008$), and real-time vintages confirm the COVID timing while downgrading 2008. Two extensions probe the edges of the protocol: an offline changepoint method (PELT), calibrated to the same false-alarm rate, matches the raw CUSUM on an obvious level break but is far weaker on variance breaks; and a bounded-memory (“MOSUM”-style) statistic fixes the raw CUSUM’s complete failure to see a second, independent level break, though the same fix does not extend to variance-type second events.

JEL classification: C12, C22, C52. Keywords: structural change; sequential change detection; CUSUM; state-space models; Kalman filter; prewhitening; false-alarm rate; latent-state diagnostics; volatility regimes.

1. Introduction

Structural change is usually sought in observables. But many economic quantities of interest — trend output, a natural rate, an underlying volatility regime — are latent states observed only with noise, and the natural first step is to filter: estimate the state, then watch the estimate for breaks. Intuition says

the filtering should help, by stripping observation noise before the change is sought. This paper asks when that intuition holds, under a protocol that makes the question answerable. Our contributions are as follows.

- Contribution 1 (protocol): a calibrated-FAR parity harness. Every method's threshold comes from the same routine on the same matched-null draws; empirical FAR is re-verified on fresh nulls; causality is enforced structurally (parameters fit on a training prefix only; forward-only filtering; alarm scores are NaN on training data) and tested bit-identically (perturbing future observations leaves all scores at earlier times unchanged, exactly). The protocol extends to the honesty of the research process itself: three pre-registered hypotheses were falsified, and every post-hoc change is logged (CHANGELOG) and the failures reported as findings rather than suppressed.
- Contribution 2 (negative + positive results): the intuition is wrong for first moments at high persistence — and we can prove why (fast-or-never theorem), then bound the claim by sweeping φ ($\mu \infty$ sorts detection, and the innovation CUSUM escapes the regime at low φ). For second moments the answer is a two-rung, two-channel decomposition. The ladder collapses to *two* rungs because the ARIMA and Kalman rungs are the same filter — the observable is exactly ARMA(1,1), an equivalence we state as theory and confirm to machine precision — so "prewhitening" and "state estimation" are not separable here; the state layer adds nothing the ARIMA residuals do not. And the prewhitening advantage is *channel-specific*: it holds for observation-noise breaks (decisive at high SNR, where the latent signal masks the noise change and raw sits at chance) but *inverts* for state-innovation breaks, where the raw variance CUSUM matches or beats the whitened rungs — the channel that matters for the volatility-regime events of the application.
- Contribution 3 (method): a tail-robust exceedance-indicator CUSUM that preserves the second-moment advantage under heavy-tailed noise, reached by rejecting two documented alternatives — a falsified clipped variant and an in-composite variant diluted below its standalone form (§8.3).
- Contribution 4 (application discipline): attribution, permutation tests, sensitivity, pinned data snapshots, and real-time vintages for the real-data claims — including a self-correction on the headline GFC timing.

Related work. The paper sits at the intersection of several literatures.

Quickest detection and SPC. Sequential change detection descends from Page's (1954) CUSUM and the quickest-detection tradition (Lorden 1971; Moustakides 1986); we use the CUSUM as the common statistic across information sets rather than proposing a new stopping rule. That literature also supplies the evaluation currency we adopt: the average run length (ARL), with methods compared at a matched in-control ARL_0 — the statistical-process-control convention (Page 1954; Montgomery 2013).

Innovation-based state-space monitoring. The idea of monitoring a *state-space* model through its innovations is not ours: it is the core of the innovation-based change-detection literature — the generalized likelihood ratio test on Kalman innovations (Willsky & Jones 1976) and the systematic treatment of CUSUM/GLR schemes on innovation sequences in Basseville & Nikiforov (1993). Our contribution on this axis is *not* the innovation CUSUM or ARL-matching per se, both of which are standard in SPC and quickest detection; it is that this matched-error protocol is essentially *absent* from the applied latent-state econometrics literature, and that transplanting it there — calibrating every information set to a common false-alarm rate — is precisely what produces the negative results (raw data wins on levels; the state estimate adds nothing over ARIMA whitening on variances).

Econometric CUSUM-of-residuals. In econometrics, sequential monitoring of structural change is the CUSUM-of-recursive-residuals line and its modern moving-window ("MOSUM") monitoring form (Brown, Durbin & Evans 1975; Chu, Stinchcombe & White 1996), which watch *observable* regression

residuals; our object is a *latent* state, and the residuals we monitor are a filter's innovations. The bounded-memory fix we give the multi-break re-arm failure in §7 is in this same MOSUM family: a moving two-window mean-shift statistic, rather than a CUSUM accumulated against a fixed training-prefix baseline.

Regime-switching. Regime-switching models (Hamilton 1989; Kim & Nelson 1999) offer a state-aware alternative whose regime probabilities we include as a benchmark and find saturate under calibration on nonstationary data.

Great Moderation and volatility empirics. The empirical target of the second-moment results is the Great Moderation volatility decline (McConnell & Pérez-Quirós 2000; Stock & Watson 2002); as we show, that and the crisis-volatility events are *state-innovation* (shock-variance) breaks, the channel on which raw and prewhitened detectors are interchangeable.

Parametric volatility models. A separate literature models conditional variance parametrically — GARCH (Bollerslev 1986) and stochastic-volatility state-space models (Kim, Shephard & Chib 1998); its object is *estimation* of a volatility process, not distribution-free monitoring at a calibrated false-alarm rate, so it is a complementary benchmark left to future work rather than a competitor here.

Offline changepoint detection. Offline changepoint methods (PELT, Killick et al. 2012) solve a retrospective segmentation problem; our monitoring is strictly causal and calibrated to a false-alarm rate, so the two are not comparable on delay (§8.5). We nonetheless calibrate PELT to the same false-alarm rate on null paths and ask whether it *localizes* a break at all, given the full sample: it is competitive with the raw-Y CUSUM on an obvious 3σ level shift (localization rate 0.83–0.92 across SNRs vs. the causal detector's 0.97–0.99) but far weaker on pure variance breaks (0.00–0.02 vs. the dedicated raw variance-CUSUM's 0.10–1.00 across the same grid), because PELT's default cost model is a mean-shift detector — a concrete reason a dedicated variance statistic, not an off-the-shelf offline method, is the right tool for the channel this paper is about.

What is new here is not any single detector but the calibrated-parity harness that makes latent-state and raw-data detectors directly comparable, and a reduced-form result (the exact ARMA(1,1) equivalence of the ARIMA and Kalman rungs) that dissolves the "does filtering help?" question for second moments into a prewhitening question with a known answer.

2. Framework and evaluation protocol

Model layer. $S_t = \phi S_{t-1} + w_t$ (var q), $Y_t = S_t + v_t$ (var r); the estimator sees only Y and fits (ϕ, q, r) by maximum likelihood on a training prefix of 125 observations (25% of the $T = 500$ baseline sample; the same 25% fraction at other T), then runs a forward-only filter with frozen parameters. Standardized one-step innovations e_t and the filtered state are the raw material for diagnostics.

Diagnostics layer. Eleven features of the filtered path — level change, slope, acceleration, instability, rolling persistence, a Page CUSUM of innovations (allowance $k = 0.5$), variance CUSUMs of $e^2 - 1$ ($k = 0.25$ and 0.05), a quietness CUSUM of $1 - e^2$ ($k = 0.05$), rolling innovation autocorrelation, and a CUSUM of the filtered state against its training baseline. Each feature is standardized at every time point by the median and IQR of its null distribution at that same time point (pooled-over-time standardization is a design flaw that blunted the composite; §8.4). The composite score is the max of standardized features; the composite is itself calibrated on nulls, which prices in the multiple-feature testing automatically.

Calibration parity. Threshold = $(1 - \text{FAR})$ quantile of the per-replication maximum score over matched-null draws; identical routine, budgets, and seed layout for LSC detectors and benchmarks

(raw-Y CUSUM, ARIMA+CUSUM, plain-HMM regime flips). Calibration, evaluation, FAR-check, and feature-scale seeds are disjoint by construction.

Metrics. Detection rate, pre-break false alarms, censored delay; for multi-break: event-level precision/recall/F1 with one-to-one greedy matching in a 100-observation window. To connect the protocol to the statistical-process-control and quickest-detection literatures we also report the two average-run-length quantities those fields are built around (`paper_assets/ar1_table.csv`). The in-control ARL_0 is the mean observations between false alarms implied by the calibrated per-observation false-alarm hazard: with monitored window length $L = 375$ ($T = 500$, 25% training) and empirical window-FAR α , the hazard is $p = 1 - (1-\alpha)^{1/L}$ and $ARL_0 = 1/p \approx L/\alpha$; our 5% window target corresponds to $ARL_0 \approx 7300$ observations, and the empirical detectors sit at 5900–7600 (raw CUSUM calibrates slightly hot). The out-of-control ARL_1 is the post-break detection delay (mean delay conditional on detection, reported beside the detection rate since misses are censored): e.g. at 3σ level, SNR 0.5, $ARL_1 \approx 77$ –86 observations. Calibrating a common ARL_0 is exactly the ARL-matching convention of SPC (Basseville & Nikiforov 1993); we express it as a window-FAR because our horizon is finite.

Arena	Method	Empirical FAR	ARL_0 (obs)
SNR 0.1	<code>lsc_composite</code>	5.2%	7023
SNR 0.1	<code>lsc_kalman_cusum</code>	3.4%	10841
SNR 0.1	<code>lsc_state_cusum</code>	5.0%	7311
SNR 0.1	<code>raw_cusum</code>	4.0%	9187
SNR 0.5	<code>lsc_composite</code>	4.8%	7624
SNR 0.5	<code>lsc_kalman_cusum</code>	4.8%	7624
SNR 0.5	<code>lsc_state_cusum</code>	5.4%	6756
SNR 0.5	<code>raw_cusum</code>	6.2%	5859
SNR 2.0	<code>lsc_composite</code>	6.2%	5859
SNR 2.0	<code>lsc_kalman_cusum</code>	4.8%	7624
SNR 2.0	<code>lsc_state_cusum</code>	7.8%	4618
SNR 2.0	<code>raw_cusum</code>	8.2%	4383

Table 1. ARL_0 by arena and method (`grid_v1` core, $T = 500$; target $ARL_0 \approx 7300$ at the 5% window-FAR).

Arena	Scenario	Method	Detect rate	ARL_1 (mean delay)
SNR 0.1	level 3σ	<code>lsc_composite</code>	0.766	112.6
SNR 0.1	level 3σ	<code>lsc_kalman_cusum</code>	0.654	63.9
SNR 0.1	level 3σ	<code>raw_cusum</code>	0.966	72.3
SNR 0.1	variance $\times 3$	<code>lsc_composite</code>	0.990	29.3
SNR 0.1	variance $\times 3$	<code>lsc_kalman_cusum</code>	0.870	80.4
SNR 0.1	variance $\times 3$	<code>raw_cusum</code>	0.728	104.0
SNR 0.5	level 3σ	<code>lsc_composite</code>	0.530	86.4
SNR 0.5	level 3σ	<code>lsc_kalman_cusum</code>	0.554	77.2
SNR 0.5	level 3σ	<code>raw_cusum</code>	0.990	81.8
SNR 0.5	variance $\times 3$	<code>lsc_composite</code>	0.992	25.2
SNR 0.5	variance $\times 3$	<code>lsc_kalman_cusum</code>	0.224	107.2
SNR 0.5	variance $\times 3$	<code>raw_cusum</code>	0.076	137.6
SNR 2.0	level 3σ	<code>lsc_composite</code>	0.670	63.2
SNR 2.0	level 3σ	<code>lsc_kalman_cusum</code>	0.674	49.0
SNR 2.0	level 3σ	<code>raw_cusum</code>	0.988	96.6
SNR 2.0	variance $\times 3$	<code>lsc_composite</code>	0.976	17.2
SNR 2.0	variance $\times 3$	<code>lsc_kalman_cusum</code>	0.268	114.6
SNR 2.0	variance $\times 3$	<code>raw_cusum</code>	0.058	172.9

Table 2. ARL_1 (detection rate and mean delay conditional on detection) at the canonical level- 3σ and variance- $\times 3$ breaks, $T = 500$.

3. Simulation design

Arenas: AR(1) latent state ($\varphi = 0.95$) at spec-SNR (stationary state variance / observation variance) $\in \{0.1, 0.5, 2.0\}$, with a persistence sweep $\varphi \in \{0.5, 0.8, 0.95, 0.99\}$ (SNR held fixed by $q = \text{SNR} \cdot (1 - \varphi^2)$) used to map the fast-or-never boundary (§4); and a local-level (random-walk state) arena, analyzed in §8.4, where — as we now demonstrate rather than assert — level detection is degenerate for every

method and variance detection becomes possible only after whitening. Breaks at mid-sample come in two variance channels whose distinction is central to §5. An *observation-noise* (r) break scales the white-noise component's standard deviation ($\times 1.5$, $\times 3$, $\times 2/3$ quieting); a *state-innovation* (q) break scales the state shock's standard deviation by the same factors — the identical SD convention, so " $\times 1.5$ " means $SD \times 1.5$ in both channels. The two are structurally different: an r -break changes only the marginal variance of Y , whereas a q -break changes both the marginal variance and the autocorrelation structure (it shifts the reduced-form ARMA(1,1) MA parameter). Also: level shifts ($0.5, 1, 3 \sigma_{\text{ref}}$), logistic ramps, and pure persistence changes ($\varphi \rightarrow 0.995$ or 0.80 with the stationary variance held fixed). $T = 500$ baseline with a $T \in \{200, 2000\}$ sweep; misspecification arenas with t_5 observation noise and a nonlinear tanh state drift.

4. First moments: raw data wins, and we can say exactly why

Empirics. At matched FAR, raw- Y CUSUM has the best level-shift detection rate at every SNR ($0.97\text{--}0.99$ at 3σ) and every T ; the latent state CUSUM approaches but never overtakes it ($0.19 \rightarrow 0.94$ as SNR rises). Two pre-registered "latent advantage" hypotheses were falsified. The latent-innovation CUSUM is the speed champion conditional on firing: median delay 24–53 observations at 3σ versus raw's 58–91, at the cost of detection rate ($0.55\text{--}0.67$) — "fast or never."

Theory. The empirics above are not an artifact of tuning; they follow from two facts about the detectors' post-break drift. Statements and proofs are in Appendix B (long-form companion: `experiments/THEORY.md`).

Proposition 1 (fast-or-never). After a state level shift δ , the standardized innovation mean decays geometrically at rate $\rho = \varphi(1-K)$ to $\mu_\infty = \delta(1-\varphi)/((1-\varphi(1-K))\sqrt{F})$, where K is the steady-state Kalman gain and F the innovation variance. If $\mu_\infty < k$, the post-transient innovation CUSUM has negative drift, and the probability of an alarm in the next L observations is bounded by $(L+1) \cdot \exp(-2(k-\mu_\infty)h)$ for threshold h .

The detector therefore fires during the adaptation transient or, with exponentially small probability, never. With $\varphi = 0.95$ and $k = 0.5$, $\mu_\infty > k$ would require shifts of order 10σ : the innovation CUSUM is *structurally* in the fast-or-never regime, which is why it is the speed champion but never the detection champion.

Fast-or-never is a property of the whitening filter, not of Kalman. By the ARMA(1,1) equivalence (§5, Appendix B) the innovations here are the innovations of the reduced-form ARMA(1,1) of Y , so Proposition 1 can be stated with no state space at all: after the shift, the standardized ARMA(1,1) innovation mean decays geometrically *at rate equal to the MA parameter* $\theta = \varphi(1-K)$ to the same $\mu_\infty = \delta(1-\varphi)/((1-\theta)\sqrt{\sigma_\varepsilon^2})$, and the bound applies verbatim. Two consequences: an ARIMA(1,0,1) residual CUSUM inherits the *same* fast-or-never behavior as the Kalman-innovation CUSUM (they whiten to the same series — nothing is Kalman-specific), and the decay rate θ is *observable* — an analyst reads it off a fitted ARMA(1,1) without positing a latent state. Since $\theta \rightarrow 1$ as $\varphi \rightarrow 1$, the trap is a persistence phenomenon, which is what the φ sweep tests next.

Proposition 2 (raw-CUSUM delay). The raw- Y CUSUM sees the full shift as a sustained standardized drift $\Delta = \delta/\sigma_Y$, giving a Wald first-passage mean delay of approximately $h/(\Delta-k)$ once $\Delta > k$ (Wald 1947; Siegmund 1985), and negligible power when $\Delta \leq k$.

Verification (`experiments/exp06_theory_check.py`, 1000 reps): the Monte Carlo innovation path matches μ_t within MC error; the Proposition 1 bound is never violated; at the actual calibrated thresh-

olds, μ_∞ sorts every observed detection rate — $\delta \leq 1\sigma$ gives bound $\leq 0.7\%$ (observed $\approx$ FAR); $\delta = 3\sigma$ is a knife-edge ($\mu_\infty = 0.43\text{--}0.48$ vs $k = 0.5$) matching the observed $0.55\text{--}0.67$. Proposition 2's Wald delays 68/84/110 across SNRs bracket the observed raw medians 58/75/91 ($\approx 15\text{--}20\%$ conservative) and explain the partial 1σ detection (0.30) without any fitting: $\Delta = 0.577$ barely exceeds k , so the Wald delay (1334) dwarfs the 250-observation horizon.

The persistence boundary (ϕ sweep). Proposition 1's regime is not unconditional — $\mu_\infty = \delta(1-\phi)/((1-\phi(1-K))\sqrt{F})$ is *increasing* in $(1-\phi)$, so the fast-or-never trap should loosen as the state becomes less persistent. Sweeping $\phi \in \{0.5, 0.8, 0.95, 0.99\}$ at fixed SNR (Figure 1) confirms it and turns the theory into a falsifiable ordering: μ_∞ sorts the innovation-CUSUM detection rate across all cells (Spearman 0.94), fast-regime cells ($\mu_\infty \geq k = 0.5$) detect $0.83\text{--}1.00$ while never-regime cells ($\mu_\infty < 0.5$) detect $0.07\text{--}0.67$, and at 3σ the detector escapes the trap at low persistence ($0.98/0.97$ at $\phi = 0.5/0.8$) but is caught at high persistence ($0.65/0.30$ at $\phi = 0.95/0.99$) — while the raw CUSUM detects at $0.96\text{--}1.00$ for *every* ϕ , so the ϕ -dependence is specific to the filtered detector. The negative first-moment result is therefore a boundary condition of *persistent* latent states — which is the empirically relevant case (trend output, a natural rate, a volatility level are near-unit-root), and the case the rest of the paper studies at $\phi = 0.95$. The one off-trend cell — $\phi = 0.99$, low SNR, where detection (0.63) exceeds what μ_∞ (0.21) predicts — is honest evidence of the transient's own contribution: at near-unit-root ϕ the adaptation transient is so long that its accumulated mass fires the CUSUM even when the asymptotic drift is negligible (the "fast" branch, strengthened by a long transient). μ_∞ governs the post-transient tail; total detection adds the transient mass.

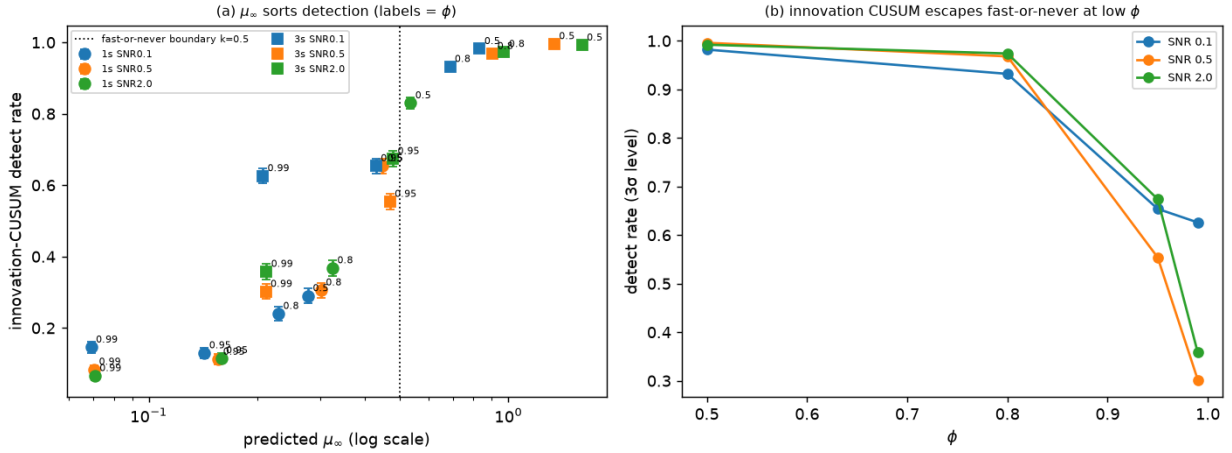


Figure 1. The ϕ sweep (grid_v6_phisweep). Left: predicted asymptotic innovation drift μ_∞ against the observed innovation-CUSUM detection rate for every $\phi \times \text{SNR} \times \text{shift}$ cell (point labels = ϕ); μ_∞ sorts detection (Spearman 0.94), and the dotted line marks the fast-or-never boundary $k = 0.5$. Right: at 3σ the detector escapes the trap at low persistence ($0.97\text{--}1.00$ at $\phi = 0.5\text{--}0.8$) and is caught at high persistence ($0.30\text{--}0.67$ at $\phi = 0.95\text{--}0.99$).

Assumptions and estimation error. Propositions 1–2 assume the steady-state filter with *known* parameters. Two facts keep this from being a limitation in practice. First, the filter reaches its steady state within the 125-observation training prefix of §2 (25% of $T = 500$), well before monitoring begins (the innovation autocorrelation is flat by then). Second, the error from estimating (ϕ, q, r) rather than knowing them is second-order: in the M1 equivalence check (§5, experiments/exp07) the ARIMA and Kalman standardized-innovation series computed with *estimated* parameters correlate at $\bar{\rho} = 0.99$ with each other and, computed with *true* parameters, at $\rho = 1.000$ to a max discrepancy of $\approx 10^{-9}$ — so the

entire estimated-vs-true gap is the small residual that leaves the median correlation at 0.99. The theory describes the estimated filter to within that gap. The scope of this claim should be stated precisely: $\bar{\rho} = 0.99$ establishes agreement of the innovation *series* under estimation; the detection-rate consequences of estimation error are shown empirically — every grid in the paper is run with estimated parameters — not proved analytically.

5. Second moments: a whitening ladder

Is the latent layer's second-moment advantage about the *state estimate*, or merely about *prewhitening* the autocorrelated observations before applying a variance statistic? We answer this directly by running the identical variance CUSUM (up-arm Page CUSUMs of $z^2 - 1$ at allowances $k = 0.25$ and 0.05 , a down-arm CUSUM of $1 - z^2$, max over arms, no per-time standardization) at three levels of prewhitening, calibrated by the same routine on the same matched-null seed blocks:

- raw — z from the raw observations, standardized by frozen training-prefix moments;
- ARIMA — the same statistic on the standardized one-step residuals of an AIC-selected, training-prefix-frozen ARIMA model (whitened, but not state-aware);
- latent — the composite's e^2 -based variance features on the Kalman innovations (state-aware).

The ladder has three rungs but only two are distinct. For the AR(1)

- noise DGP the observable Y has an *exact* ARMA(1,1) reduced form: differencing by $(1 - \phi L)$ leaves an MA(1), whose invertible root gives an MA parameter θ and innovation variance σ_ϵ^2 satisfying two identities we verify to machine precision (Appendix B; `lsc.theory.arma11_representation`) — $\sigma_\epsilon^2 = F$, the Kalman innovation variance, and $\theta = \rho = \phi(1 - K)$, the Proposition-1 decay rate. So the steady-state Kalman innovations *are* the ARMA(1,1) innovations: the ARIMA and latent rungs are the same filter, not two competitors. We confirm this numerically (`experiments/exp07_arma_equivalence.py`, ≥ 200 null paths): with true parameters the two standardized-innovation series agree to a median correlation of 1.000 (max discrepancy $\approx 10^{-9}$), and with each rung fit on the training prefix — the actual operating condition — they still correlate at $\bar{\rho} = 0.99$, the small wedge being pure estimation error (forcing the ARIMA order to the true (1,0,1) tightens it to 0.9995). (AIC, incidentally, rarely selects (1,0,1): near the unit root at $\phi = 0.95$ it prefers (1,0,0) or a differencing (0,1,1); this is a benign near-unit-root artifact — those orders approximate the ARMA(1,1) closely enough to preserve $\bar{\rho} \geq 0.95$ — reported in full in Appendix B.) The practical consequence: the ladder is really raw vs. whitened, and “does the *state estimate* help beyond ARIMA whitening?” has the answer *no, by construction*. What remains is the genuinely empirical question — when does whitening help at all? — which turns out to depend on *which variance channel* breaks.

The ladder, both channels (detection rate at $T = 500$, 5% calibrated FAR; MC SEs ≤ 0.02 in `paper_assets/ladder_table.csv`, `break_channel` column).

channel	break	rung	SNR 0.1	SNR 0.5	SNR 2.0
r (obs-noise)	$\times 1.5$	raw	1.00	0.56	0.10
		ARIMA	0.90	0.94	0.87
		latent	0.82	0.87	0.91
	$\times 3$	raw	1.00	1.00	0.85
		ARIMA	0.98	1.00	1.00
		latent	0.99	0.99	0.98
q (state-innov)	$\times 1.5$	raw	0.09	0.21	0.23
		ARIMA	0.03	0.10	0.16
		latent	0.06	0.11	0.23
	$\times 3$	raw	0.72	0.96	0.96
		ARIMA	0.26	0.79	1.00
		latent	0.44	0.76	0.98

channel	break	rung	SNR 0.1	SNR 0.5	SNR 2.0
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Reading the ladder: the ordering inverts across channels. The subtle $\times 1.5$ break is the discriminating case, and it tells opposite stories in the two channels.

Observation-noise (r) breaks — prewhitening wins. The raw rung is strongly SNR-dependent, falling monotonically from 0.996 (SNR 0.1) through 0.560 to 0.102 (SNR 2.0, within 5 pp of the 6.0% empirical FAR, i.e. chance). The mechanism is transparent: as SNR rises the latent state's variance dominates the marginal variance of Y, so a $\times 1.5$ change in the (now-small) observation-noise component is a shrinking fraction of total variance, masked by state-driven autocorrelation. Prewhitening removes exactly that autocorrelation: the ARIMA rung is flat across SNR (0.90 / 0.94 / 0.87) and tracks the *latent* rung step for step (0.82 / 0.87 / 0.91) — as it must, since the equivalence above makes them the same filter. On this channel the advantage over raw is *prewhitening under autocorrelation*, decisive where the latent signal masks the noise change.

State-innovation (q) breaks — the ordering inverts, read separately by break size. The coarse $\times 3$ break carries the inversion claim cleanly: every rung detects above chance, and the raw rung matches or beats the whitened rungs at every SNR (raw 0.72 / 0.96 / 0.96 vs ARIMA 0.26 / 0.79 / 1.00, the whitened rung catching up only at the high-SNR ceiling). The *subtle* $\times 1.5$ break must be read against the same ≤ 5 -pp-from-FAR "chance" standard applied to the r channel above (empirical FARs on this grid are 4.2–6.6%): raw at SNR 0.1 (0.09) and ARIMA at SNR 0.1 and 0.5 (0.03, 0.10) are within 5 pp of their empirical FARs — chance — and only the higher-SNR cells clear the standard (raw 0.21 / 0.23 at SNR 0.5 / 2.0; ARIMA 0.16 at SNR 2.0). The honest subtle-break statement is therefore not "raw detects it" but "whitening fails to recover it": no rung detects a $\times 1.5$ shock-variance break well, and whitening only loses ground. The raw rung's SNR-dependence still *reverses sign* relative to the r channel — it rises with SNR (0.09 $\rightarrow$ 0.21 $\rightarrow$ 0.23) — and the mechanism is the mirror image: a q-break inflates the *state's own* variance, which dominates the marginal variance of Y at high SNR, so a raw z^2 statistic sees it directly — whereas prewhitening *strips out* the state-carried signal along with the autocorrelation. Prewhitening reveals a break in the white component and removes a break in the state. Quieting ($\times 2/3$, i.e. reduced q) is undetectable by every rung (≤ 0.07 , at FAR): a low-q state contributes too little to Y to register its own reduction.

Where does raw's q-break advantage come from? A ϕ sweep. Holding the shock variance q and observation variance r fixed and sweeping the persistence ϕ makes the state's stationary variance $q/(1-\phi^2)$ — and hence the induced SNR — rise as the $1/(1-\phi^2)$ amplification factor (Figure 2; configs/grid_v8_phiqbbreak.yaml). The result is instructive and only partly what one might guess. On the *subtle* $\times 1.5$ break, raw's advantage $\Delta = \text{detect}(\text{raw}) - \text{detect}(\text{ARIMA})$ is driven by exactly this amplification: it vanishes when the state is white ($\Delta = 0.00$ at $\phi = 0.1$, where whitening is a no-op), rises with the amplification (Spearman 0.83), and — the clean check — equals the SNR-sweep value from the fixed- ϕ grid at the *same induced SNR* ($\Delta = 0.11$ vs 0.11 at SNR 0.5; 0.07 vs 0.07 at SNR ≈ 2), so the ϕ -sweep and the SNR-sweep are one experiment: raw's edge on a subtle shock-variance break is the state's variance share, nothing more. (These Δ values are differences between near-floor detection rates — e.g. 0.21 vs 0.10 at SNR 0.5 — so they measure the visibility of a marginal advantage, not a large power gap.) But the law is not monotone to the unit root — Δ peaks at $\phi = 0.95$ and recedes at $\phi = 0.99$, because there the raw detector's own baseline degrades (its calibrated threshold jumps from 275 to 1829, the nonstationarity penalty of §8.4). And on the *coarse* $\times 3$ break the amplification story does not hold at all: raw beats ARIMA at every ϕ , including $\phi = 0.1$ ($\Delta = 0.34$), with Spearman(amp, Δ) negative — a gross shock-variance break is visible to raw z^2 regardless of amplification, while

whitening removes it. (This is the whitening itself, not order mis-selection: at low ϕ the ARIMA rung selects a stationary AR(1), not a differencing model.) The honest reading: prewhitening strips the state-carried variance signal for q -breaks of any size — that is B2 — and the $1/(1-\phi^2)$ amplification governs only *how visible* the residual raw advantage is on the marginal, subtle break.

ϕ	Amplification $1/(1-\phi^2)$	Δ , subtle $\times 1.5$ (raw – ARIMA)	Δ , coarse $\times 3$ (raw – ARIMA)
0.10	1.01	0.000	0.344
0.50	1.33	0.016	0.526
0.70	1.96	0.038	0.528
0.85	3.60	0.104	0.204
0.95	10.26	0.112	0.168
0.99	50.25	0.074	0.302

Table 3. Raw’s detection-rate advantage over the ARIMA rung (Δ), swept over ϕ at fixed q , r (*grid_v8_phiqbreak*). On the subtle break Δ tracks the amplification factor and peaks at $\phi = 0.95$ before receding at the unit-root edge; on the coarse break Δ stays large at every ϕ , including $\phi = 0.1$ where amplification is negligible.

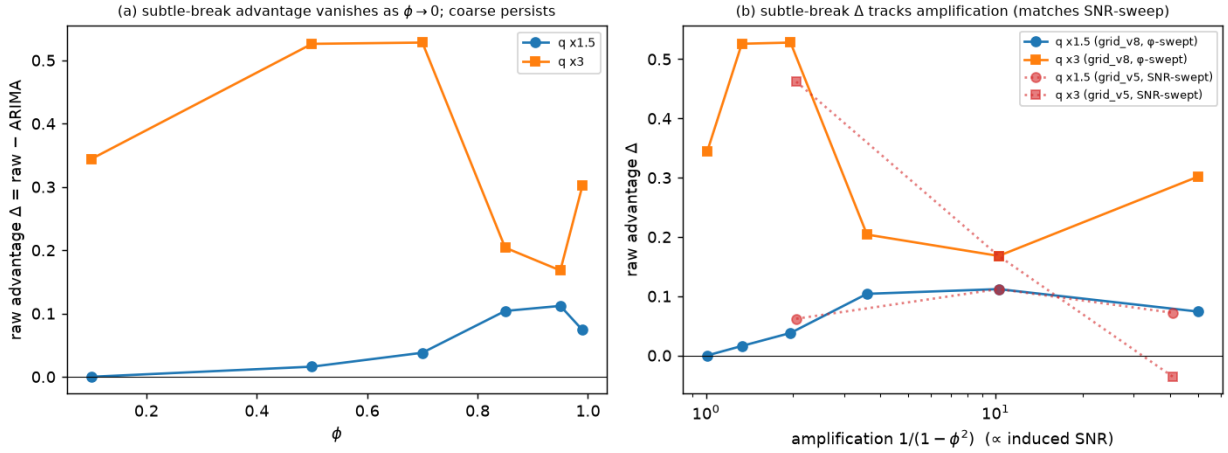


Figure 2. The $\phi \times q$ cross-grid (*grid_v8_phiqbreak*). Left: raw’s advantage $\Delta = \text{detect}(\text{raw}) - \text{detect}(\text{ARIMA})$ against ϕ ; the subtle $\times 1.5$ advantage vanishes at $\phi = 0.1$ and peaks at $\phi = 0.95$, while the coarse $\times 3$ advantage persists at every ϕ . Right: the same Δ against the $1/(1-\phi^2)$ amplification factor, with the fixed- ϕ SNR-sweep values (*grid_v5*) overlaid — on the subtle break the ϕ -swept and SNR-swept experiments coincide at matched induced SNR; note the Δ values are differences between near-floor detection rates.

This resolves the pre-registered decision rule (*experiments/CHANGELOG.md*) as Outcome B2: the “prewhitening beats raw” result is *specific to the observation-noise channel*. It matters because the events the application targets — the Great Moderation, crisis volatility — are state-innovation (shock-variance) breaks, not observation-noise breaks; on that channel a raw variance CUSUM is at least as good as any amount of whitening — decisively on the coarse break, while on the subtle break every rung is near the floor and whitening only loses ground. This is not a weakness to hide but the explanation for a real-data fact (§9): the raw variance CUSUM’s crisis timing is *indistinguishable* from the state-aware composite’s, exactly as the q -channel predicts. (The earlier r -channel-only framing recorded a provisional “Outcome C”; the q channel is what disambiguates it.) Attribution: the z^2/e^2 variance-pressure arms drive the alarms at every rung.

Runway (T sweep, $\times 1.5$, SNR 0.5). The subtle-break detection scales with sample length for every rung:

at $T = 200$ all three are weak (raw 0.26, ARIMA 0.10, latent composite 0.11) — the whitened rungs need runway to accumulate the CUSUM — and by $T = 2000$ all reach ceiling (raw 0.98, ARIMA ≈ 1.00 , latent 0.99). The $T = 200$ calibration is *not* hot for these variance detectors (empirical FAR 5.0–5.8% vs the 5% target), unlike the level CUSUMs (§8.1). At the coarse $\times 3$ break the composite is at ceiling even at $T = 200$ (1.00, median delay 26 of the ~ 100 post-break observations, best level-oriented benchmark 0.17).

Scale *quieting* ($\times \frac{2}{3}$) inverts the up-break pattern and is treated in §6 and §8.3: there the raw rung catches the low-SNR case (0.32 at SNR 0.1, where noise dominates) while the whitened rungs and composite sit at chance, and only the standalone exceedance detector recovers it at moderate SNR.

6. Dynamics: near the information floor

Pure persistence changes (marginals preserved) are close to undetectable: at SNR 0.5, no method exceeds FAR (a pre-registered hypothesis wrong in both directions — raw CUSUM even reached 0.16 on persistence-up via a conditional “level-freeze” artifact we dissect). Quieting changes (φ down) actively *suppress* every excursion statistic below its null level. Purpose-built quietness features (CUSUM of $1 - e^2$, rolling innovation autocorrelation) rescue detection only where the information exists: 0.33 at SNR 2.0 (the only above-FAR persistence detection anywhere in the grid) and 0.17 at $T = 2000$. That 0.33 is a single favorable cell, not a broad capability: its value is that it maps where the information floor sits, showing detection is possible exactly where the state is most visible and nowhere else. Scale-quieting ($\times \frac{2}{3}$) is detectable — but, outside the noise-dominated regime, only by the exceedance detector (§8.3): at SNR 0.5 it reaches 0.41/0.33 (Gaussian/ t_5) while every other method sits at chance. The one exception is the same low-SNR window that lifts the raw rung on the up-breaks (§5): at SNR 0.1 the raw variance CUSUM catches the quieting too (0.32), because when observation noise dominates a scale change is directly visible in raw z^2 regardless of its sign.

7. Multiple breaks: everyone is one-shot

Under a re-arm protocol applied identically to all methods (re-arm when the score drains below half threshold plus a 20-obs refractory), raw CUSUM never detects a second event (recall 0.00 — its fixed-baseline statistic saturates and cannot drain), but the LSC CUSUMs rarely re-arm in time either at 150-observation spacing (level $\rightarrow$ level second-event recall ≤ 0.05): first-alarm delay plus drain time exceeds the gap. The exception is cross-channel pairs: level $\rightarrow$ variance is caught by the composite alone (second-event recall 0.60, F1 0.63) because its variance features were never saturated by the level event. Re-arming costs almost nothing under the null ($\leq 1.2\%$ of null paths give a second alarm) — saturation, not chatter, binds.

A bounded-memory fix, and its limits (2026-07-16). The diagnosis above — a fixed-baseline statistic compares every observation against the *original* training-prefix reference and so never drains after a permanent shift — points to a specific repair: replace the fixed reference with a moving one. We build a MOSUM-style two-window statistic (the same family as the CUSUM-of-recursive-residuals monitoring literature; Chu, Stinchcombe & White 1996) that compares a trailing window’s mean to the window immediately before it, rather than to the training baseline, for both the raw-Y and innovation-CUSUM channels (windowed_break_pressure). On the level $\rightarrow$ level scenario this closes the gap dramatically: the windowed raw-Y statistic’s second-event recall rises from 0.004 to 0.682 — essentially matching its own first-event recall (0.692) — at *higher* precision than the unwindowed statistic (0.99 vs 0.80), a real fix rather than a threshold trick, at a modest first-event recall cost (0.692 vs 0.738). The windowed

innovation-CUSUM improves less (second-event recall $0.008 \rightarrow 0.234$) because the filter's own adaptivity (μ_∞ , Proposition 1) already partially "forgets" a level shift, leaving less room for a moving reference to add. The fix is channel-specific, not general: on level $\rightarrow$ variance and variance $\rightarrow$ variance scenarios, both windowed statistics stay at second-event recall ≈ 0.00 , because they are mean-shift statistics and a pure variance change carries no mean signal for a moving-window *mean* comparison to see — closing that gap needs a windowed *variance* statistic (a moving-window analogue of the r/q-channel CUSUMs of §5), left to future work. Multi-break detection therefore needs the statistic's channel matched to the break, exactly as the single-break results of §5 already required — the two sections are the same lesson at different time scales, and the one real economic setting where it bites is a recession cluster (§9): a level shock followed by a shock-variance regime change is exactly the level $\rightarrow$ variance case the composite already handles, but two level shocks in close succession — a double-dip — are exactly where every fixed-baseline statistic here still fails.

8. Robustness

We stress-test the §5–§7 results along five axes: sample size, distributional misspecification, a repaired heavy-tailed statistic, composite-vs-standalone dilution, and real-data sensitivity to the false-alarm target and window length.

8.1 Sample size. §5 numbers; short-T caveat: heavy-tailed null maxima make quantile thresholds noisy, and at $T = 200$ the two baseline CUSUMs calibrate hot (8.8–9.4% vs the 5% target). Quote empirical FARs alongside any short-sample power claim.

8.2 Misspecification. t_5 observation noise: rankings preserved; raw CUSUM unhurt on levels (0.99); the composite's subtle-variance case collapses ($\times 1.5$: $0.87 \rightarrow 0.16$) — repaired in §8.3. The raw and ARIMA variance rungs are far less tail-fragile than the composite: on the $\times 1.5$ break at SNR 0.5 the raw rung falls only $0.56 \rightarrow 0.43$ and the ARIMA rung $0.94 \rightarrow 0.74$ under t_5 (against the composite's $0.87 \rightarrow 0.16$), because the plain max-over-arms z^2/e^2 statistic without per-time composite standardization retains the tail excursions that carry the variance signal — the same logic that motivates the standalone exceedance detector (§8.3). At $\times 3$ all variance rungs stay at ceiling under t_5 (raw and ARIMA 1.00, composite 0.97). Nonlinear tanh drift (mildly bimodal state): level detection collapses for *every* method (raw 0.15 at 3σ — the state's own regime-hopping inflates all null thresholds), while variance detection is untouched (0.97 at $\times 1.5$). The two detection families read disjoint information channels.

8.3 Heavy tails and the exceedance repair (a three-act story worth telling honestly). (i) Huberizing e^2 (clip at $2.5 \cdot \text{MAD}$) — falsified, worse everywhere, even Gaussian ($\times 1.5$: $0.87 \rightarrow 0.06$): the variance signal lives in the tail the clip removes. (ii) An exceedance-indicator CUSUM (count of $|e|$ above its training 90th percentile; bounded summand under any distribution) — the raw statistic separates the null and break classes near-perfectly. Dropped into the composite in place of the e^2 -pressure features (the "robust2" variant), and standardized per-time-point as in §8.4, it is *diluted, not dead*: it reaches 0.58 at $\times 1.5$ (SNR 0.5, Gaussian) and 0.21 at $\times 1.5$ under t_5 — where it even edges the e^2 -based composite's heavy-tail collapse (0.16, §8.2) — and ≈ 0.97 at $\times 3$ under t_5 . But it trails the standalone detector of (iii) at the subtle break, because a max-over- ≈ 10 -features composite spreads the calibrated 5% false-alarm budget across all features: the exceedance feature's per-t standardized score climbs past $z \approx 19$ on a $\times 1.5$ break, so bounded increments *do* reach discriminating ratios (an earlier pooled-scale build, before the §8.4 fix, wrongly suggested otherwise), yet the shared threshold still costs it power relative to calibrating the statistic alone. The remedy is *exposure*, not a different feature. (iii) The same statistic as a *standalone* calibrated detector (up-arm $k = 0.05$, down-arm $k = 0.02$; k chosen on non-evaluation seeds, procedure logged): variance $\times 1.5$ at 0.87 Gaussian / 0.75 t_5 (repairing 0.16), $\times 3$ at ~ 1.0 with ~ 37 -

obs delay under both distributions, and the first successful quieting detection ($\times^{2/3}$: 0.41/0.33). Both headline rates fell 3–5pp short of the pre-registered bars — reported as such.

8.4 The local-level (random-walk state) arena: degenerate for levels, whitening-mandatory for variances. The canonical latent-state model in economics is the local level (random-walk state); we ran the level and variance ladder cells there rather than dismiss it (Figure 3; `configs/grid_v7_l1level.yaml`), and it splits cleanly. *Level* detection is degenerate for every method: at 3σ all five detectors sit at the 5% FAR (raw CUSUM 0.07–0.10, Kalman-innovation 0.04–0.15, raw/ARIMA variance and composite ≤ 0.07). A level break in a random-walk state is absorbed by a well-specified filter as one large ordinary innovation — no sustained signal — and the raw-Y CUSUM has no fixed baseline: its calibrated threshold is 1500–2400 (versus $O(10-100)$ in the AR(1) arena) and it still calibrates hot. There is no common null against which to rank level detectors, which is why we study the identifiable AR(1) arena in the body. But *variance* detection is not degenerate — it merely requires whitening. On a $\times 1.5$ observation-noise break the raw z^2 CUSUM is at chance (0.06 at every SNR; threshold $\approx 10^4$, meaningless on a nonstationary series) while the ARIMA-differencing rung detects (0.71 / 0.84 / 0.58) and the Kalman composite detects (0.97 / 0.89 / 0.68); at $\times 3$ the whitened rungs reach 0.97–1.00 while raw stays at chance. This is the exact complement of §5's r-channel AR(1) result — where a raw variance CUSUM could *win* when observation noise dominates — and it sharpens the paper's thesis: prewhitening is not merely helpful but *mandatory* once the observable is nonstationary, because the raw statistic has no stationary baseline to calibrate against.

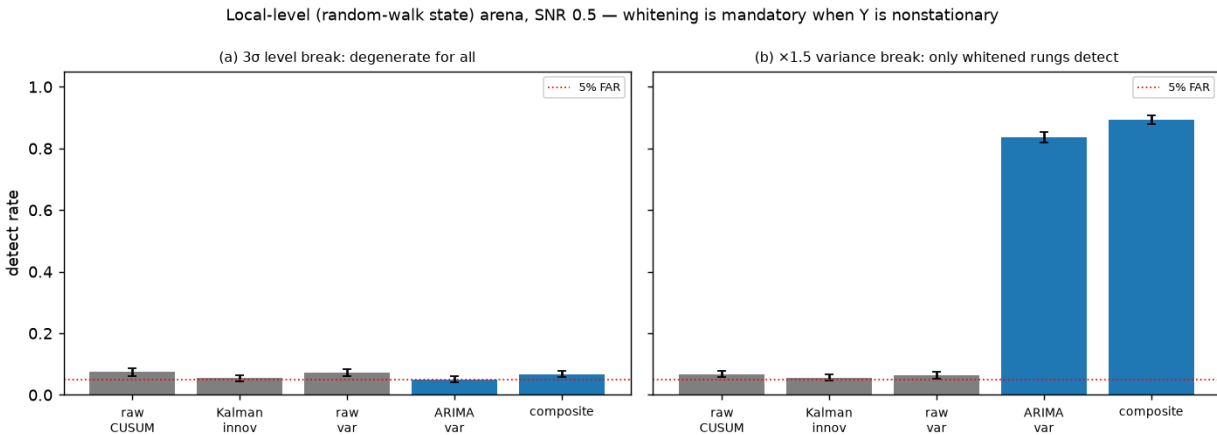


Figure 3. The local-level (random-walk state) arena at SNR 0.5 (`grid_v7_l1level`). Left: a 3σ level break is degenerate for every method — all five detectors sit at the 5% FAR line. Right: a $\times 1.5$ observation-noise variance break is caught only by the whitened rungs (ARIMA-differencing 0.84, Kalman composite 0.89) while the raw variance CUSUM stays at chance — whitening is mandatory once Y is nonstationary.

8.5 An offline benchmark: PELT, calibrated to the same false-alarm rate (2026-07-16). Related work dismisses offline changepoint methods as solving a different problem (retrospective segmentation, not causal monitoring), which is true but does not by itself say how well an off-the-shelf method would do if pressed into the same role. We calibrate PELT (Killick, Fearnhead & Eckley 2012; ruptures, l2 cost, applied to Y standardized by the training-prefix mean/std — the raw rung's own standardization) to a 5% false-alarm rate on null AR(1) paths by bisecting its penalty parameter, exactly mirroring the threshold-calibration protocol used for every causal detector, then ask whether it *localizes* a break within 25 observations, given the full sample — an offline-localization question, deliberately not a delay

comparison, since PELT sees future data the causal detectors cannot.

Arena (SNR)	Scenario	PELT localize rate
0.1	level 0.5σ	0.02
0.1	level 1σ	0.07
0.1	level 3σ	0.83
0.1	variance $\times 1.5$	0.00
0.1	variance $\times 3$	0.20
0.5	level 0.5σ	0.03
0.5	level 1σ	0.11
0.5	level 3σ	0.91
0.5	variance $\times 1.5$	0.01
0.5	variance $\times 3$	0.01
2.0	level 0.5σ	0.02
2.0	level 1σ	0.11
2.0	level 3σ	0.92
2.0	variance $\times 1.5$	0.02
2.0	variance $\times 3$	0.02

Table 4. PELT localization rate at a FAR-matched (5%) operating point, $n = 300$ per cell (*exp08_pelt*).

On the canonical 3σ level break PELT is competitive with the causal raw-Y CUSUM (0.83–0.92 here vs. 0.97–0.99 in Table 1’s arena, seeing the whole path rather than detecting online) — an off-the-shelf offline method genuinely can find an obvious level shift. But on variance breaks it is close to useless (0.00–0.02, against the dedicated raw variance-CUSUM’s ladder-table numbers of 0.10–1.00, §5), because PELT’s default l2 cost model is a mean-shift statistic: it responds to a variance change only through the second-order effect that within-segment sum-of-squares grows, a far weaker signal than a statistic built for variance directly. The comparison sharpens rather than undercuts the paper’s thesis: an off-the-shelf offline segmentation method is not a substitute for a channel-matched statistic, even when it is given the unfair advantage of seeing the whole sample.

Protocol lessons (each cost us a wrong result before it was fixed). plain-HMM regime probabilities saturate and cannot be FAR-calibrated on nonstationary data; probability-scale scores need log-odds; EM needs persistent-initialization restarts; composite features must be standardized per-time-point, not pooled; order-statistic thresholds have Beta($n+1-k$, k) noise regardless of distribution, so heavy-tailed detectors need larger calibration budgets.

9. Real data (illustrative)

Four FRED series — industrial production, GDP, and 10-year Treasury yields (pinned snapshots, 2026-07-11) plus the unemployment rate (pinned 2026-07-16) — rolling causal monitoring (train 120 months / monitor 60), per-segment parametric bootstrap calibration at 5% FAR per window, alarms attributed to the feature that crossed. Throughout this section the detectors are distribution-free monitors calibrated to a false-alarm rate; no parametric conditional-variance model is estimated and no such claim is made. Association with NBER-registered events is tested by permutation: the observed count of registered events “hit” (an alarm within 12 months after) is compared to the distribution of hit counts from 20,000 resamples of the same number of alarm months drawn uniformly from all monitored months; the reported p is the fraction of resamples at least as extreme as the observed count, so a small p means alarms cluster after registered events more than chance alone would produce (*real_data_eval.py*).

Series	Method	Alarms	Hits / events	Stray	Perm. p
INDPRO	lsc_composite	4	3/9	1	0.008
INDPRO	lsc_kalman_cusum	2	2/9	0	0.021
INDPRO	lsc_tail_cusum	5	0/9	5	1.000
INDPRO	raw_cusum	1	1/9	0	0.148
INDPRO	raw_var_cusum	5	1/9	4	0.554
GDP	lsc_composite	3	2/9	1	0.067
GDP	lsc_kalman_cusum	1	1/9	0	0.164

Series	Method	Alarms	Hits / events	Stray	Perm. p
GDP	lsc_tail_cusum	3	2/9	1	0.063
GDP	raw_cusum	0	—	—	—
GDP	raw_var_cusum	2	1/9	1	0.309
GS10	lsc_composite	7	1/3	6	0.325
GS10	lsc_kalman_cusum	2	1/3	1	0.109
GS10	lsc_tail_cusum	4	0/3	4	1.000
GS10	raw_cusum	2	1/3	1	0.104
GS10	raw_var_cusum	9	1/3	8	0.402
UNRATE	lsc_composite	3	2/9	1	0.057
UNRATE	lsc_kalman_cusum	4	4/9	0	0.0002
UNRATE	lsc_tail_cusum	6	2/9	4	0.205
UNRATE	raw_cusum	4	4/9	0	0.0004
UNRATE	raw_var_cusum	7	2/9	5	0.257

Table 5. Real-data alarm summary at 5% FAR, 120-month training (*rd_eval.csv*). GDP's *raw_cusum* fired zero alarms across all 12 windows, so no permutation test applies. UNRATE (new, P2) is the single strongest association in the table — both *raw_cusum* and *lsc_kalman_cusum* fire exactly 4 alarms, all 4 landing on NBER peaks (zero strays), out of 9 peaks in range.

Industrial production (INDPRO, 1948–2026). Composite alarms: 2008-09 and 2020-04 (both variance_pressure), 1990-12 (variance_quiet), 1969-08 (variance_quiet; within the false-alarm budget and reported as such). Hits 3/9 NBER peaks within 12 months, 1 stray vs 0.7 expected; permutation $p = 0.008$ (innovation CUSUM $p = 0.021$; raw CUSUM 1 hit, $p = 0.15$). The raw variance CUSUM — the bottom rung of the ladder, added here to test real-data uniqueness — does catch the GFC (2008-09, up-arm, the same month as the composite) but embeds it among four stray quieting alarms (1967, 1968, 1988, 2019) and misses COVID, so its NBER association is not significant (5 alarms, 1 hit, $p = 0.55$). The crisis is detectable by a raw variance statistic; what the latent layer buys on real data is a *clean* alarm profile — a significant, low-stray association — not the crisis catch itself. A permutation-test implementation note: the p-values in this section are recomputed with a per-series, per-method seed (2026-07-16) after we found the original shared, sequentially-consumed random generator made one series' p-value depend on which *other* series happened to be present in the run — harmless for any single number in isolation, but a reproducibility hazard the fix removes; every value moved by ≤ 0.002 from the originally reported estimates, consistent with Monte Carlo noise at 20,000 draws, not a change in any underlying alarm.

What real-time data changes (ALFRED vintages). Re-running each decision on the data *as it existed that month* both tempers and sharpens the timing claims. COVID: robust — the composite's real-time alarm is at the 2020-04 vintage (data month 2020-03), one month earlier than on revised data and ~2 months before the NBER announcement. GFC: downgraded — the crossing is at the same data month (2008-09) but only in the 2008-12 vintage, so real-time knowledge is coincident with the NBER announcement (2008-12-01), though still 4 months ahead of the raw level CUSUM's real-time alarm (2009-04). Because the raw variance CUSUM alarmed on revised INDPRO, we ran it through the identical vintage protocol for an apples-to-apples comparison: its real-time timing is *indistinguishable from the composite's* — same vintage and same data month for both GFC (2008-12 vintage, 2008-09) and COVID (2020-04 vintage, 2020-03), both ahead of the raw level CUSUM. On real-data crisis *timing*, then, the prewhitening rung and the state-aware composite are interchangeable; the composite's advantage is confined to the clean association profile of §9 and to the simulation-calibrated $\times 1.5$ subtlety threshold of §5, not to real-time crisis detection.

GDP (GDPC1, quarterly). GFC and COVID caught (variance_pressure, 2008Q4 and 2020Q2). The raw variance CUSUM again catches both crises (2009Q2 and 2020Q2, up-arm) with fewer alarms than on INDPRO (2 alarms, 1 hit, $p = 0.31$) — the same "catches the crisis, misses the clean association" pattern. The registered Great Moderation event (1984Q1) is an honest miss: causal rolling monitoring detects the quieting only in 1992Q2 (composite) / 1997Q1 (tail shortfall), because 60-quarter training windows

contain 1970s volatility until the early 1990s. Retrospective full-sample break dates are not reproducible by honest monitoring.

10-year Treasury yield changes (GS10). Volcker regime caught 4 months after the registered 1979-10 event (variance_pressure; raw and innovation CUSUM the same month); the post-Volcker disinflation quietings are flagged (1989–1996, shortfall/variance_quiet); the 2008 ZLB event is missed within 12 months. A third registered event, the 2022-03 hiking-cycle onset, is missed by *every* method within 24 months (rd_gs10_summary.csv) — unlike Volcker, a persistent directional yield rise without a matching shock-variance regime shift falls outside every detector's reach here, a genuine miss reported as such rather than a horizon-window artifact. The raw variance CUSUM also flags Volcker (1980-02, up-arm) but is the noisiest detector on this series (9 alarms, 8 stray, 1/3 events hit, $p = 0.40$) — on a series that is *all* volatility regime, a raw z^2 statistic fires on every regime shift, which is exactly why its event-association washes out.

Sensitivity. All numbers below are on INDPRO, the headline series.

Variant	Method	Alarms	Hits / events	Perm. p
FAR 1%	lsc_composite	3	3/9	0.003
FAR 1%	lsc_kalman_cusum	1	1/9	0.146
FAR 1%	lsc_tail_cusum	1	0/9	1.000
FAR 1%	raw_cusum	1	0/9	1.000
FAR 1%	raw_var_cusum	4	2/9	0.100
FAR 5% (baseline)	lsc_composite	4	3/9	0.008
FAR 5% (baseline)	lsc_kalman_cusum	2	2/9	0.021
FAR 5% (baseline)	lsc_tail_cusum	5	0/9	1.000
FAR 5% (baseline)	raw_cusum	1	1/9	0.148
FAR 5% (baseline)	raw_var_cusum	5	1/9	0.554
FAR 10%	lsc_composite	5	2/9	0.148
FAR 10%	lsc_kalman_cusum	3	2/9	0.055
FAR 10%	lsc_tail_cusum	5	2/9	0.148
FAR 10%	raw_cusum	2	2/9	0.018
FAR 10%	raw_var_cusum	6	1/9	0.619
FAR 20%	lsc_composite	9	1/9	0.773
FAR 20%	lsc_kalman_cusum	3	2/9	0.052
FAR 20%	lsc_tail_cusum	7	1/9	0.682
FAR 20%	raw_cusum	2	2/9	0.020
FAR 20%	raw_var_cusum	8	1/9	0.727
Window 180 mo	lsc_composite	14	2/8	0.556
Window 180 mo	lsc_kalman_cusum	3	2/8	0.046
Window 180 mo	lsc_tail_cusum	5	3/8	0.015
Window 180 mo	raw_cusum	2	2/8	0.017
Window 180 mo	raw_var_cusum	14	1/8	0.878

Table 6. INDPRO sensitivity to the false-alarm target (1/5/10/20%, 120-month training) and to a longer training window (180 months, 5% FAR).

Two independent stress tests both isolate the composite's association as fragile *relative to itself*, not to the other detectors. The FAR sweep is monotone and clean for the composite: tightening from 5% to 1% *sharpens* the association ($p = 0.008 \rightarrow 0.003$, alarm count falls from 4 to 3 as the marginal, least-reliable alarm drops out), while relaxing to 10% and 20% floods it with noise ($p = 0.148$, then 0.773) — exactly the dilution a well-behaved calibrated statistic should show. raw_cusum and lsc_kalman_cusum stay comparatively stable across the same sweep (p in the 0.02–0.15 range throughout), because they fire too few alarms for the FAR target to matter much either way. Training window 180 months breaks both variance-based detectors' bootstrap calibration on nonstationary real data — the composite and the raw variance CUSUM each fire 14 alarms / 21 windows ($p = 0.556$ and 0.878, both uninformative) — while the level (raw CUSUM), innovation, and tail detectors stay sane (2–5 alarms, p as low as 0.015–0.046). It is the *second-moment* statistic, not the composite machinery, that is sensitive to a training window long enough to straddle a volatility regime; training windows must be short enough to be locally stationary (120 months worked).

10. Discussion

Four points summarize what the harness bought us.

- The latent-state diagnostics layer is not a better level-shift detector; it is a *different instrument*, reading second moments and dynamics. But the whitening ladder (§5) sharpens the claim past the point where “the state estimate” survives as an ingredient: on this DGP the Kalman innovations *are* the ARMA(1,1) innovations (an exact reduced-form equivalence, verified to machine precision), so whatever second-moment power the latent layer has is prewhitening, full stop — ARIMA residuals capture it identically. What the state framing still buys is dynamics features (persistence, quieting) and the fast-or-never speed edge on levels. The division of labor is now measured (grids), reduced (raw vs. whitened, the two whitened rungs proven identical), and derived (fast-or-never with its ϕ boundary).
- Practical recipe, now channel-aware. Run raw CUSUM for levels. For scale changes the right move depends on *which variance moves*, and one usually does not know: (i) if the break is in the observation (measurement) noise, *whiten first* (ARIMA residuals suffice; the Kalman state adds nothing) — prewhitening is decisive when the series is autocorrelated and the latent signal is strong, and a raw variance CUSUM works only when observation noise dominates; (ii) if the break is in the state’s own shock variance — the Great-Moderation / crisis- volatility case — a raw variance CUSUM is at least as good as whitening at every SNR, and whitening can *hurt*. Because the practitioner rarely knows the channel, running *both* a raw and a whitened variance CUSUM (cheap, and they agree on the q channel) is the robust default. Use the exceedance-indicator variant under heavy tails and the composite for breadth. Calibrate everything on matched nulls at a common FAR (a common ARL_0) and report empirical FARs.
- The calibrated-parity protocol itself is a contribution: it exposed every failure mode above, and it is what makes the negative results informative rather than anecdotal.
- Limitations / future work: a bounded-memory (MOSUM-style) statistic fixes the multiple-breaks re-arm failure for level-type second events (§7) but not variance-type ones — a windowed *variance* statistic is the natural next step; adaptive composite weighting (breadth tax); switching-SSM (Kim filter) model layer; formalizing the persistence-break mechanisms; a vol-regime reference set for scoring the exceedance detector on real data; and benchmarking against parametric volatility models (GARCH, stochastic volatility) — a different object, conditional-variance estimation rather than distribution-free monitoring, but the natural complementary comparison for the second-moment results.

What, then, is the latent layer actually for? The results invite a deflationary reading, and we take it seriously rather than deflect it. For raw *detection power*, the state-space layer is largely redundant: raw CUSUM owns level shifts (§4); ARIMA whitening owns observation-noise variance and, by the exact ARMA(1,1) equivalence, *is* the Kalman innovation filter — the state estimate contributes nothing beyond it (§5); and on the shock-variance channel that motivates the application, a raw variance CUSUM matches or beats whitening outright (§5, whose ϕ sweep traces the residual advantage to the state’s $1/(1-\phi^2)$ variance amplification on subtle breaks). A practitioner armed with a raw CUSUM, a raw variance CUSUM, and an off-the-shelf ARIMA residual CUSUM would reproduce most of the detection frontier without ever writing down a latent state. So what survives as genuinely the latent layer’s? Four things, and only these. (1) *Dynamics, at the floor*. Persistence and quieting changes have no raw analogue; the state features (rolling innovation autocorrelation, the $1-e^2$ quietness CUSUM) are the *only* above-FAR detection of a pure persistence change anywhere in the grid — but that detection is a single favorable cell (0.33 at SNR 2.0), a capability at the information floor rather than a robust one. (2) *Speed*. Conditional on firing, the innovation CUSUM is the fastest level detector (median delay 24–53 vs

raw's 58–91) — the very "fast" half of fast-or-never. (3) *A single attributable instrument*. The composite reads ~10 channels under one calibrated FAR budget and reports *which* crossed, which is what turns a real-data alarm into an interpretable event (every crisis alarm attributes to a named second-moment feature, §9). (4) *A clean association profile*. On real data the composite's crisis *timing* is no better than a raw variance CUSUM's, but its *selectivity* is — a significant, low-stray association with NBER dates ($p = 0.008$ vs 0.55) that a raw z^2 statistic firing on every wiggle cannot match. The honest one-sentence answer: the latent layer is not a better detector of any single break type; it is a *breadth-and-interpretation* instrument whose irreducible content is dynamics (weakly) and attribution (robustly), not the second-moment power one might have expected filtering to buy.

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Appendix A. Reproducibility

`make all` regenerates every table and figure from pinned seeds (Python 3.14, statsmodels/hmmlearn; `make fred` / `make realdata` / `make realtime` for the data applications, snapshots under `data/`). The referee-hardening round added six reproducible artifacts to the pack: `exp07` (ARMA equivalence), `grid_v5` (the q-break channel), `grid_v6` (the ϕ sweep), `grid_v7` (the local-level arena), `grid_v8` (the $\phi \times q$ amplification cross-grid), and `arl` (ARL₀/ARL₁ table); all are pinned-seed and join the existing grids draw-for-draw. A second round (2026-07-16) added `exp08` (the PELT localization benchmark, §8.5) and extended `exp04` with the two windowed-CUSUM methods (§7) and `realdata` with a fourth series (unemployment, `unrate`), a third GS10 event (the 2022 hiking cycle), and a false-alarm-rate sweep (1%, 5%, 10%, 20%) on INDPRO. 98 tests include bit-identical no-lookahead checks for every feature and detector (including the raw and ARIMA variance rungs, the two windowed statistics, and a training-freeze check), DGP ground-truth checks (including the state-innovation break's SD scaling and null-match), the ARMA(1,1)/Riccati identity and Kalman-equivalence guards, calibration-parity checks, composite golden-score regression guards (which pin the per-time-point-standardized output so a stale artifact cannot recur), and the windowed-statistic bounded-memory (drain-after-permanent-shift) property. All post-hoc design changes and pre-registered hypotheses (including the three falsified ones and the two rejected robust-feature designs — one falsified, one diluted, §8.3) are in `experiments/CHANGELOG.md`; full experiment narratives in `experiments/FINDINGS.md`; theory statements and proofs in Appendix B, with `experiments/THEORY.md` as the long-form companion. The complete source, pinned data snapshots, and seed configuration constitute the replication package, to be posted publicly on acceptance and available from the author on request in the interim.

Appendix B. Theory: statements and proofs

This appendix gives self-contained statements and proofs of Propositions 1–2; `experiments/THEORY.md` remains the long-form companion, and the numerical verification is in `experiments/exp06_theory_check.py` (1000 replications, §4).

Setup and assumptions. Scalar state-space model with *known* parameters, filter in steady state, Gaussian innovations: $S_t = \phi S_{t-1} + w_t$ with $w_t \sim N(0, q)$ and $|\phi| < 1$, and $Y_t = S_t + v_t$ with $v_t \sim N(0, r)$. The steady-state prediction variance P solves the Riccati fixed point $P = \phi^2 P r / (P + r) + q$; the gain is $K = P / (P + r)$ and the innovation variance $F = P + r$. Standardized one-step innovations $e_t = (Y_t - \phi \hat{S}_{t-1}) / \sqrt{F}$ are iid $N(0, 1)$ under the null. A level break adds δ to the state path from t_0 on: $\tilde{S}_t = S_t + \delta \cdot 1\{t \geq t_0\}$, hence $\tilde{Y}_t = Y_t + \delta \cdot 1\{t \geq t_0\}$ — the DGP used in all experiments. The one-sided Page CUSUM with drift allowance k and threshold h is $g_t = \max(0, g_{t-1} + e_t - k)$, alarming when $g_t \geq h$. (The experiments use fitted training-prefix parameters, diffuse initialization, and a two-sided CUSUM; §4 explains why the known-parameter theory nevertheless describes them to first order.)

Proposition 1 (fast-or-never; restated from §4). After a state level shift δ at t_0 : (a) the broken path's standardized innovations decompose as $\tilde{e}_t = e_t + \mu_t$ with e_t the null innovations and μ_t deterministic, decaying geometrically at rate $\rho = \phi(1-K) \in (0, 1)$ from μ_{t_0}

$= \delta/\sqrt{F}$ to $\mu_\infty = \delta(1-\varphi)/((1-\varphi(1-K))\sqrt{F})$; (b) if $\mu_t \leq \tilde{\mu} < k$ for all $t \geq t_1 \geq t_0$ (post-transient) and $g_{\{t_1\}} = g < h$, then for any horizon L , $P(\max_{t_1 < t \leq t_1+L} g_t \geq h \mid g_{\{t_1\}} = g) \leq (L+1) \cdot \exp(-2(k-\tilde{\mu})(h-g))$.

Proof of (a). The steady-state filter is a linear time-invariant map of Y , so the innovations of the broken path decompose as the null innovations plus the deterministic innovation response μ_t to the input $\delta \cdot 1_{\{t \geq t_0\}}$. Write a_j for the filter's mean state-estimate response j steps after the break, $a_j = E[\hat{S}_{\{t_0+j\}}] - E[\hat{S}_{\{t_0+j\}}^0]$. The filter predicts $\varphi a_{\{j-1\}}$ and corrects by K times the mean innovation: $a_j = \varphi a_{\{j-1\}} + K(\delta - \varphi a_{\{j-1\}}) = \rho a_{\{j-1\}} + K\delta$ with $a_{\{-1\}} = 0$ and $\rho = \varphi(1-K)$; the mean innovation response is the input minus the prediction response, $\mu_{\{t_0+j\}} = (\delta - \varphi a_{\{j-1\}})/\sqrt{F}$. Solving the linear recursion, $a_j = (K\delta/(1-\rho))(1 - \rho^{j+1})$, so μ decays geometrically at rate ρ from $\delta/\sqrt{F}$ to the limit $(\delta - \varphi K\delta/(1-\rho))/\sqrt{F} = \delta(1-\varphi)/((1-\varphi(1-K))\sqrt{F}) = \mu_\infty$. ■

Proof of (b). For $n > t_1$, $g_n = \max(g + \sum_{i=t_1+1}^n X_i, \max_{t_1 < m \leq n} \sum_{i=m}^n X_i)$ with increments $X_i = e_i + \mu_i - k = z_i - (k - \mu_i)$, z_i iid $N(0,1)$. An alarm by t_1+L requires some anchored sum $\sum_{i=m}^n X_i \geq h - g$ for an anchor $m \in (t_1, t_1+L]$ (or the g -anchored sum $\geq h - g$). Each X_i is stochastically dominated by $z_i - (k - \tilde{\mu})$, for which $\theta^* = 2(k - \tilde{\mu})$ solves $E[e^{\theta^* X}] = 1$; the exponential martingale $e^{\theta^* \sum}$ with the maximal inequality gives $P(\sup_n \sum_{i=m}^n X_i \geq h - g) \leq e^{-\theta^*(h-g)}$ for each of the $\leq L+1$ anchor points, and a union bound finishes. ■

Interpretation: the filter adapts, so of the full shift δ only the fraction $(1-\varphi)/(1-\varphi(1-K))$ survives in the innovations per step. The transient carries total excess mass $\sum_j (\mu_{\{t_0+j\}} - \mu_\infty) = \varphi a_\infty / ((1-\rho)\sqrt{F})$, which is what a "fast" detection consumes; if the alarm does not fire on the transient, part (b) says it fires later with probability exponentially small in the threshold — fast or never.

Proposition 2 (raw-CUSUM delay: Wald approximation; restated from §4). The raw- Y CUSUM standardizes Y by its frozen training moments; after the break the standardized mean shift $\Delta = \delta/\sigma_Y$ ($\sigma_Y^2 = q/(1-\varphi^2) + r$) persists for all $t \geq t_0$. If $\Delta > k$ the post-break increments have positive drift $\Delta - k$ and the alarm is certain as the horizon grows, with first-passage (Wald) mean delay $E[D] \approx h/(\Delta - k)$; if $\Delta \leq k$, the bound of Proposition 1(b) applies with $\tilde{\mu} = \Delta$.

Derivation. Unlike the innovations, the raw standardized series has no adaptation: the shift δ enters Y permanently and the training moments are frozen, so every post-break increment is $z_i + \Delta - k$ with drift $\Delta - k > 0$. The CUSUM then behaves as a positive-drift random walk, and Wald's identity for the first passage of level $h - g$ gives $E[D] \approx (h - g)/(\Delta - k) \approx h/(\Delta - k)$ (Wald 1947). The approximation ignores boundary overshoot and reflection at 0, so it is an approximation, not a bound (a corrected version is in Siegmund 1985); against the grids it runs ≈ 15 –20% conservative at 3σ (§4). If $\Delta \leq k$ the drift is nonpositive and the never-detect bound applies verbatim. ■

ARMA(1,1) equivalence of the whitened rungs (§5). Applying the AR operator to the observable, $(1 - \varphi L)Y_t = w_t + v_t - \varphi v_{\{t-1\}} =: u_t$, which is an MA(1) with autocovariances $\gamma_u(0) = q + r(1 + \varphi^2)$, $\gamma_u(1) = -\varphi r$, $\gamma_u(h) = 0$ ($h \geq 2$). Matching u_t to $(1 - \theta L)\varepsilon_t$ (variance σ_ε^2) gives the invertible root $\theta = (m - \sqrt{m^2 - 4})/2$ with $m = (q + r(1+\varphi^2))/(\varphi r)$, and $\sigma_\varepsilon^2 = \varphi r/\theta$. Two identities close the loop with Proposition 1: $\sigma_\varepsilon^2 = F$ and $\theta = \rho = \varphi(1 - K)$. The second follows from the innovation recursion $v_t = Y_t - \varphi Y_{\{t-1\}} + \varphi(1-K)v_{\{t-1\}}$ (substitute $\hat{S}_{\{t-1\}} = \varphi \hat{S}_{\{t-2\}} + K v_{\{t-1\}}$ into $v_t = Y_t - \varphi \hat{S}_{\{t-1\}}$), which is the ARMA(1,1) innovation recursion once $\theta = \varphi(1-K)$. Hence the steady-state Kalman innovations and the ARMA(1,1) innovations are the same series; the state layer is not a distinct information set from ARIMA whitening on this DGP. Both identities hold to $< 10^{-12}$ (lsc.theory.arma11_representation, test_arma11_riccati_identities),

and the numerical equivalence on null paths — with true and with estimated parameters — is in `experiments/exp07_arma_equivalence.py` (§5). A near-unit-root caveat: AIC over the benchmark order grid rarely selects the exact (1,0,1) at $\varphi = 0.95$ (it prefers (1,0,0) at low SNR, the differencing (0,1,1) at higher SNR), but these approximate the ARMA(1,1) closely enough that the median innovation correlation with the Kalman filter stays at 0.99; forcing (1,0,1) restores 0.9995.

Appendix C. Summary of key quantities

Claim	Number	Source
Raw CUSUM level 3σ detect, all SNRs	0.97–0.99	grid_v1
Innovation CUSUM delay vs raw, 3σ	24–53 vs 58–91 obs	exp02
Innovation CUSUM detect at 3σ	0.55–0.67	exp02/grid_v1
$\mu\infty$ at 3σ , SNR 0.5 (knife-edge vs $k=0.5$)	0.469	exp06
Composite variance $\times 1.5$, $T=500$	0.82 / 0.87 / 0.91 (SNR 0.1/0.5/2.0)	grid_v1
Ladder $\times 1.5$: raw rung, $T=500$	1.00 / 0.56 / 0.10 (SNR 0.1/0.5/2.0)	grid_v4_varbench
Ladder $\times 1.5$: ARIMA rung, $T=500$	0.90 / 0.94 / 0.87 (SNR 0.1/0.5/2.0)	grid_v4_varbench
Ladder $\times 1.5$ t_5 (raw/ARIMA/composite), SNR 0.5	0.43 / 0.74 / 0.16	grid_v4_varbench, v2_misspec
ARMA=Kalman innovation $\hat{p}$ (estimated / true params)	0.99 / 1.000 ($\max \Delta \approx 10^{-9}$)	exp07
q-break $\times 1.5$: raw rung, $T=500$	0.09 / 0.21 / 0.23 (SNR 0.1/0.5/2.0)	grid_v5_qbreak
q-break $\times 1.5$: ARIMA rung, $T=500$	0.03 / 0.10 / 0.16 (SNR 0.1/0.5/2.0)	grid_v5_qbreak
Pre-registered decision rule (§5) resolved	Outcome B2 (r-channel-specific)	§5, CHANGELOG
φ sweep: $\mu\infty$ sorts innovation-CUSUM detection	Spearman 0.94	grid_v6_phisweep
Innovation CUSUM 3σ escape ($\varphi=0.5$ / 0.99)	0.98–1.00 / 0.30–0.63	grid_v6_phisweep
Local-level 3σ level detect (all methods)	≤ 0.15 ($\approx$ FAR)	grid_v7_llevel
Local-level $\times 1.5$ variance (raw / ARIMA rung)	0.06 / 0.58–0.84	grid_v7_llevel
ARL ₀ at 5% window-FAR ($L=375$)	≈ 7300 obs	arl_table
$\varphi \times q$: $\times 1.5$ raw edge, φ -swept = SNR-swept	$\Delta 0.11=0.11$ (SNR 0.5)	grid_v8 vs grid_v5
$\varphi \times q$: subtle Δ at $\varphi \rightarrow 0$ / Spearman(amp, Δ)	0.00 / 0.83; $\times 3$ falsified (-0.60)	grid_v8_phiqbreak
Variance $\times 1.5$ across $T = 200/500/2000$	0.11 / 0.87 / 0.99	grid_v2_T
t_5 collapse and repair ($\times 1.5$)	0.16 $\rightarrow$ 0.75 (tail_cusum)	grid_v2_misspec, v3c
Quieting $\times 2/3$ (only tail_cusum)	0.41 / 0.33	grid_v3c
Persistence-down, best anywhere	0.33 (SNR 2.0)	grid_v1
Multi-break: raw second-event recall (level $\rightarrow$ level)	0.00	exp04
Composite level $\rightarrow$ var second event	0.60 (F1 0.63)	exp04
Windowed-CUSUM fix, level $\rightarrow$ level 2nd event (raw / innovation)	0.00 $\rightarrow$ 0.68 / 0.01 $\rightarrow$ 0.23	exp04
Windowed-CUSUM fix, level $\rightarrow$ var / var $\rightarrow$ var 2nd event	no improvement (≈ 0.00): mean-shift only	exp04
PELT localization at FAR-matched 5%, level 3σ	0.83–0.92 (vs. causal raw CUSUM 0.97–0.99)	exp08_pelt
PELT localization at FAR-matched 5%, variance $\times 1.5/\times 3$	0.00–0.20 (vs. dedicated raw variance-CUSUM 0.10–1.00)	exp08_pelt
INDPRO permutation p (composite)	0.008	rd_eval
GFC real-time	2008-09 data, known 2008-12	rd_realtime
COVID real-time	data 2020-03, ~ 2 mo before NBER	rd_realtime